Supplement to "Identification and inference for the shape of a distribution from rounded reports"

Equation, theorem and section numbers without the prefix S refer to the main text. Section S1 proves Theorem 2; Sections S2–S5 prove the results on the reporting mechanism and state and prove Proposition S1 on the band estimator; Section S6 collects additional simulation results.

S1 Proof of Theorem 2

S1.1 Conventions

Throughout, $[\ell] = \{1, \ldots, \ell\}$, $\mathcal{S}$ is the simplex of $\mathbb{R}^K$, $\mathbf{1} = (1, \ldots, 1)$, $H = \{x \in \mathbb{R}^K : \sum_i x_i = 0\}$, and $y(C) = \sum_{i \in C} y_i$ for $y \in \mathbb{R}^\ell$ and $C \subseteq [\ell]$. The *sign pattern* of $p \in \mathbb{R}^K$ is $\sigma(p) = (\mathrm{sgn}(p_{k+1} - p_k))_{k<K} \in \{-1, 0, 1\}^{K-1}$; $p \in \mathcal{U}$ if and only if no -1 precedes a $+1$ in $\sigma(p)$. A *run* of p is a maximal set of consecutive cells on which p is constant; the runs are determined by $\sigma(p)$. If $p \in \mathcal{U}$, the cells at which p attains its maximum form a single run, the *top run*; a run to its left is a *left run*, and the run adjacent to it on the right is higher; a run to its right is a *right run*. Which runs are left, top and right is again determined by $\sigma(p)$. We read the definition of a contact with the convention $p_0 = p_{K+1} = -\infty$: $\mathcal{T}(p)$ is the set of pairs $(k, k+1)$ with $p_k = p_{k+1}$, except the pair forming the top run when the top run has exactly two cells, whether or not that run touches an end of $[K]$.

For $k \in \{1, 2\}$ let $\mathcal{C}_k(\ell)$ be the family of subsets of $[\ell]$ that are unions of at most k sets of consecutive integers, the empty set included; let

$$\mathcal{C}_k^{\rightarrow}(\ell) = \big\{L \subseteq [\ell] : \ L \cup \{\ell + 1\} \in \mathcal{C}_k(\ell + 1)\big\},$$
$$\mathcal{C}_k^{\leftarrow}(\ell) = \big\{R \subseteq [\ell] : \ \{0\} \cup R \in \mathcal{C}_k(\{0, \ldots, \ell\})\big\},$$

where $\mathcal{C}_k(\{0, \ldots, \ell\})$ is the family of unions of at most k sets of consecutive elements of $\{0, \ldots, \ell\}$. Each of the three families contains $\emptyset$ and $[\ell]$, and $\mathcal{C}_1 \subseteq \mathcal{C}_2$ in each case. For a pair

$\mathcal{C} = (\mathcal{C}_1, \mathcal{C}_2)$ of this kind, $y \in \mathbb{R}^\ell$ and $\mu \in \mathbb{R}$ put

$$E_k^{\mathcal{C}}(y,\mu) = \max_{C \in \mathcal{C}_k} \{y(C) - |C|\mu\}, \qquad D^{\mathcal{C}}(y,\mu) = E_2^{\mathcal{C}}(y,\mu) - E_1^{\mathcal{C}}(y,\mu) \geq 0,$$

$$\Delta^{\mathcal{C}}(y) = \sup_{\mu \in \mathbb{R}} D^{\mathcal{C}}(y,\mu),$$

and write Δ, $\Delta^{\rightarrow}$, $\Delta^{\leftarrow}$ (and E_k, $E_k^{\rightarrow}$, $E_k^{\leftarrow}$, D, $D^{\rightarrow}$, $D^{\leftarrow}$) for the three families. If $y \in \mathcal{S}$ and $\mu < 0$ then every $y_i - \mu$ is positive, both maxima are attained at $[\ell]$ and $D(y,\mu) = 0$; so on $\mathcal{S}$ the function Δ is the statistic of the main text and the restriction $\lambda \geq 0$ there is void. Since $\Delta^{\leftarrow}(y) = \Delta^{\rightarrow}(y_\ell, \ldots, y_1)$, every statement about $\Delta^{\rightarrow}$ has a mirror image for $\Delta^{\leftarrow}$, which we do not spell out.

The data are $\hat{p} = N/n$ with $N \sim \mathrm{Mult}(n,p)$; $\mathbb{P}^*$, E^* and $\mathcal{L}^*$ denote conditional probability, expectation and law given $\hat{p}$. The bootstrap centre $\bar{p}_n$ is a function of $\hat{p}$ and n, and $\hat{p}^* = N^*/n$ with $N^* \sim \mathrm{Mult}(n, \bar{p}_n)$ given $\hat{p}$. Let $\Sigma(p) = \mathrm{diag}(p) - pp^\top$. The critical value is $c_n^* = \inf\{c : \mathbb{P}^*(\Delta(\hat{p}^*) \leq c) \geq 1 - \alpha\}$, which is ≥ 0 because $\Delta \geq 0$, and the test rejects if and only if $\Delta(\hat{p}) > c_n^*$. For laws P, Q on a Euclidean space let $d_{\mathrm{BL}}(P,Q) = \sup\{|\int h\,dP - \int h\,dQ| : \|h\|_\infty + \mathrm{Lip}(h) \leq 1\}$, which metrises weak convergence. "The conditional law of X_n^* converges in probability to $\mathcal{L}(X)$" means $d_{\mathrm{BL}}(\mathcal{L}^*(X_n^*), \mathcal{L}(X)) \xrightarrow{p} 0$.

S1.2 The functional

Lemma S1. *Let $\mathcal{C}$ be one of the three families on $[\ell]$ and, for $y \in \mathbb{R}^\ell$,*

$$M(y) = \Big\{ \frac{y(C) - y(C')}{|C| - |C'|} : \; C, C' \in \mathcal{C}_2, \; |C| \neq |C'| \Big\}.$$

(i) $\Delta^{\mathcal{C}}(y) = \max_{\mu \in M(y)} D^{\mathcal{C}}(y,\mu)$.

(ii) $\Delta^{\mathcal{C}}(ty) = t\Delta^{\mathcal{C}}(y)$ *for* $t \geq 0$, *and* $\Delta^{\mathcal{C}}(y + c\mathbf{1}) = \Delta^{\mathcal{C}}(y)$ *for* $c \in \mathbb{R}$.

(iii) $|\Delta^{\mathcal{C}}(y) - \Delta^{\mathcal{C}}(y')| \leq 2\|y - y'\|_1$.

(iv) There are finitely many closed polyhedral cones $Q_1, \ldots, Q_M$ with $\bigcup_j Q_j = \mathbb{R}^\ell$ and vectors $g_1, \ldots, g_M$ such that $\Delta^{\mathcal{C}}(y) = g_j^\top y$ for $y \in Q_j$.

Proof. (i) For fixed y, $E_k^{\mathcal{C}}(y,\cdot)$ is the upper envelope of the affine functions $\mu \mapsto y(C) - |C|\mu$, $C \in \mathcal{C}_k$. At a point where its slope changes, two of these functions with different slopes are both active, so the point belongs to $M(y)$; as $\mathcal{C}_1 \subseteq \mathcal{C}_2$ this holds for $k = 1$ and $k = 2$. Hence $D^{\mathcal{C}}(y,\cdot)$ is continuous and affine on each interval between consecutive points of the finite set $M(y)$, on $(-\infty, \min M(y)]$ and on $[\max M(y), \infty)$; $M(y) \neq \emptyset$ because $\emptyset, [\ell] \in \mathcal{C}_2$.

For $\mu < \min_i y_i$ all $y_i - \mu$ are positive, the maximum over both families is at $[\ell] \in \mathcal{C}_1$, and $D^{\mathcal{C}}(y,\mu) = 0$; for $\mu > \max_i y_i$ it is at $\emptyset \in \mathcal{C}_1$ and $D^{\mathcal{C}}(y,\mu) = 0$. An affine function on a half-line that vanishes on a sub-half-line vanishes identically, so $D^{\mathcal{C}}(y,\cdot) = 0$ outside $[\min M(y), \max M(y)]$, and the supremum of $D^{\mathcal{C}}(y,\cdot)$ is attained at a point of $M(y)$.

(ii) $E_k^{\mathcal{C}}(ty, t\mu) = tE_k^{\mathcal{C}}(y,\mu)$ for $t > 0$ and $E_k^{\mathcal{C}}(y + c\mathbf{1}, \mu + c) = E_k^{\mathcal{C}}(y,\mu)$; take suprema over μ.

(iii) $|y(C) - y'(C)| \le \|y - y'\|_1$ gives $|E_k^{\mathcal{C}}(y,\mu) - E_k^{\mathcal{C}}(y',\mu)| \le \|y - y'\|_1$, hence $|D^{\mathcal{C}}(y,\mu) - D^{\mathcal{C}}(y',\mu)| \le 2\|y - y'\|_1$ for every μ.

(iv) By (i), $\Delta^{\mathcal{C}}(y) = \max_{(C_0,C_0')} \big[\max_{C\in\mathcal{C}_2} \lambda_{C_0,C_0',C}(y) - \max_{C'\in\mathcal{C}_1} \lambda_{C_0,C_0',C'}(y)\big]$, the outer maximum over pairs in $\mathcal{C}_2$ with $|C_0| \ne |C_0'|$, where $\lambda_{C_0,C_0',C}(y) = y(C) - |C|\,(y(C_0) - y(C_0'))/(|C_0| - |C_0'|)$ is linear in y. Writing the bracket as $\max_C \min_{C'}(\lambda_{C_0,C_0',C} - \lambda_{C_0,C_0',C'})$ exhibits $\Delta^{\mathcal{C}}$ as $\max_a \min_b \ell_{a,b}$ for finitely many linear forms $\ell_{a,b}$. The sign vectors $(\operatorname{sgn}(\ell_{a,b} - \ell_{a',b'})(y))_{a,b,a',b'}$ partition $\mathbb{R}^\ell$ into finitely many relatively open polyhedral cones, on each of which the maximising a and minimising b can be chosen fixed, so that $\Delta^{\mathcal{C}}$ is linear there; by continuity the same linear function represents $\Delta^{\mathcal{C}}$ on the closure. □

Lemma S2. *(i) $\Delta(y) = 0$ if and only if $y \in \mathcal{U}$; in particular $\Delta \equiv 0$ on $\mathbb{R}^\ell$ if and only if $\ell \le 2$. (ii) $\Delta^{\to}(y) = 0$ if and only if $y_1 \le \cdots \le y_\ell$; in particular $\Delta^{\to} \equiv 0$ on $\mathbb{R}^\ell$ if and only if $\ell = 1$.*

Proof. (i) If $y \in \mathcal{U}$ then for every μ the set $\{i : y_i > \mu\}$ is a set of consecutive integers, so it belongs to $\mathcal{C}_1$, and it maximises $y(C) - |C|\mu$ over all $C \subseteq [\ell]$; hence $E_1 = E_2$. If $y \notin \mathcal{U}$ there are $i < j < k$ with $y_j < \min(y_i, y_k)$; let $y_j < \mu < \min(y_i, y_k)$ and let $L \in \mathcal{C}_1$ attain $E_1(y,\mu)$. If $i, k \in L$ then $j \in L$ and $L \setminus \{j\} \in \mathcal{C}_2$ has value $E_1(y,\mu) + (\mu - y_j)$; if $i \notin L$ then $L \cup \{i\} \in \mathcal{C}_2$ has value $E_1(y,\mu) + (y_i - \mu)$; likewise for k. In each case $D(y,\mu) > 0$. Every $y \in \mathbb{R}^\ell$ with $\ell \le 2$ is unimodal, and $(1,0,1)$ is not.

(ii) If y is nondecreasing, $\{i : y_i > \mu\}$ is a final segment of $[\ell]$, belongs to $\mathcal{C}_1^{\to}$, and maximises $y(C) - |C|\mu$ over all $C \subseteq [\ell]$, so $E_1^{\to} = E_2^{\to}$. Otherwise let $y_i > y_{i+1}$, $\mu = (y_i + y_{i+1})/2$, and let $L \in \mathcal{C}_1^{\to}$ attain $E_1^{\to}(y,\mu)$; L is a final segment of $[\ell]$, possibly empty. If $i \in L$ then $i+1 \in L$ and $L \setminus \{i+1\}$, whose union with $\{\ell+1\}$ is a union of two sets of consecutive integers, belongs to $\mathcal{C}_2^{\to}$ and has value $E_1^{\to}(y,\mu) + (\mu - y_{i+1})$. If $i \notin L$ then $L \cup \{i\} \in \mathcal{C}_2^{\to}$ has value $E_1^{\to}(y,\mu) + (y_i - \mu)$. In both cases $D^{\to}(y,\mu) > 0$. Every $y \in \mathbb{R}^1$ is nondecreasing and $(1,0)$ is not. □

For $p \in \mathcal{U}$ with distinct values $a_1 < \cdots < a_s$ let $\delta(p) = \min_{j<s}(a_{j+1} - a_j)$ ($= \infty$ if $s = 1$) and $\eta(p) = \delta(p)/(4K)$. To each run R of p attach the function $\Phi_R : \mathbb{R}^R \to [0,\infty)$ given by $\Phi_R = \Delta$ (on $\mathbb{R}^{|R|}$) if R is the top run, $\Phi_R = \Delta^{\to}$ if R is a left run and $\Phi_R = \Delta^{\leftarrow}$ if R is a

right run, and define

$$\Delta'_p(x) = \max_R \Phi_R(x_R), \qquad x \in \mathbb{R}^K, \tag{S1}$$

the maximum over the runs of p, x_R denoting the restriction of x to R in increasing order of the cells.

Lemma S3. *Let $p \in \mathcal{U}$. For every $x \in \mathbb{R}^K$ with $\|x\|_\infty < \eta(p)$,*

$$\Delta(p + x) = \Delta'_p(x).$$

The function Δ'_p is continuous, positively homogeneous, invariant under $x \mapsto x + c\mathbf{1}$, satisfies $|\Delta'_p(x) - \Delta'_p(x')| \le 2\|x - x'\|_1$, is linear on each of finitely many closed polyhedral cones covering $\mathbb{R}^K$, and depends on p only through $\sigma(p)$. The runs R with $\Phi_R \not\equiv 0$ are exactly the vertex sets of the connected components of the graph on $[K]$ with edge set $\mathcal{T}(p)$, so that the maximum in (S1) *may be restricted to those components, and $\Delta'_p \equiv 0$ if and only if $\mathcal{T}(p) = \emptyset$.*

Proof. Write $\eta = \eta(p)$, $\delta = \delta(p)$, $q = p + x$, and for $j \le s$ let $A_j = \{i : p_i > a_j\}$, $B_j = \{i : p_i = a_j\}$, $C_j = \{i : p_i < a_j\}$. Since $p \in \mathcal{U}$, the upper level sets A_j and $A_j \cup B_j$ are sets of consecutive integers; hence for $j < s$, $B_j = B_j^- \cup B_j^+$ with B_j^- the run of p immediately to the left of A_j and B_j^+ the run immediately to its right, either possibly empty, and B_j^- is a left run, B_j^+ a right run; $A_s = \emptyset$ and B_s is the top run. Every run of p occurs exactly once in this list. Fix $\lambda \in \mathbb{R}$ and put $s_i = q_i - \lambda$.

Step 1: $|\lambda - a_j| \ge \eta$ for all j. Then $|p_i - \lambda| \ge \eta > |x_i|$ for every i, so s_i is nonzero with the sign of $p_i - \lambda$ and $\{i : s_i > 0\} = \{i : p_i > \lambda\}$, a set of consecutive integers. It belongs to $\mathcal{C}_1(K)$ and maximises $\sum_{i \in C} s_i$ over all $C \subseteq [K]$, whence $D(q, \lambda) = 0$.

Step 2: $\lambda = a_j + \mu$ with $|\mu| < \eta$. Since $2\eta < \delta$, such j is unique. For $i \in A_j$, $s_i \ge \delta - 2\eta$; for $i \in C_j$, $s_i \le -(\delta - 2\eta) < 0$; for $i \in B_j$, $s_i = x_i - \mu$ and $|s_i| < 2\eta$. We claim that every $C \in \mathcal{C}_2(K)$ attaining $E_k(q, \lambda)$, $k \in \{1, 2\}$, satisfies $A_j \subseteq C \subseteq A_j \cup B_j$. If C contained a cell of C_j, then, since C_j lies outside the set of consecutive integers $A_j \cup B_j$, the smallest or the largest element of C would lie in C_j; removing it leaves a member of $\mathcal{C}_k(K)$ with a strictly larger sum. If some $a \in A_j$ is not in C: when $C = \emptyset$, the set $A_j \in \mathcal{C}_1(K)$ has sum $\ge |A_j|(\delta - 2\eta) > 0$; otherwise let G be the set of integers strictly between a and C together with a itself, that is, $G = \{a, \dots, \min C - 1\}$ if $a < \min C$, $G = \{\max C + 1, \dots, a\}$ if $a > \max C$, and G the maximal set of consecutive integers containing a and disjoint from C if $\min C < a < \max C$, in which case C consists of two sets of consecutive integers and G is the gap between them. In every case $C \cup G \in \mathcal{C}_k(K)$ and $G \subseteq A_j \cup B_j$, since $A_j \cup B_j$ is a set of consecutive integers containing a and C. As $a \in G$ and the other elements of G are

in $A_j \cup B_j$,

$$\sum_{i\in G} s_i \;\geq\; (\delta - 2\eta) - 2\eta(|G|-1) \;\geq\; \delta - 2K\eta \;\geq\; \delta/2 \;>0,$$

contradicting the optimality of C. The claim follows.

Suppose $j < s$. A set C with $A_j \subseteq C \subseteq A_j \cup B_j$ is $C = L \cup A_j \cup R$ with $L \subseteq B_j^-$, $R \subseteq B_j^+$. Identify B_j^- with $[\ell^-]$ in increasing order, so that A_j is adjacent to the cell ℓ^-, and B_j^+ with $[\ell^+]$, so that A_j is adjacent to the cell 1. The maximal sets of consecutive integers composing C are those of $L \cup \{\ell^- + 1\}$ and those of $\{0\} \cup R$, glued along A_j, so their number is $\nu(L \cup \{\ell^- + 1\}) + \nu(\{0\} \cup R) - 1$, ν counting the pieces. Hence $C \in \mathcal{C}_1(K)$ iff $L \in \mathcal{C}_1^{\rightarrow}(\ell^-)$ and $R \in \mathcal{C}_1^{\leftarrow}(\ell^+)$, and $C \in \mathcal{C}_2(K)$ iff $L \in \mathcal{C}_2^{\rightarrow}(\ell^-)$ and $R \in \mathcal{C}_1^{\leftarrow}(\ell^+)$, or $L \in \mathcal{C}_1^{\rightarrow}(\ell^-)$ and $R \in \mathcal{C}_2^{\leftarrow}(\ell^+)$. Since $s_i = x_i - \mu$ on B_j, the claim gives

$$\begin{aligned}
E_1(q,\lambda) - s(A_j) &= E_1^{\rightarrow}(x_{B_j^-},\mu) + E_1^{\leftarrow}(x_{B_j^+},\mu),\\
E_2(q,\lambda) - s(A_j) &= \max\bigl\{E_2^{\rightarrow}(x_{B_j^-},\mu) + E_1^{\leftarrow}(x_{B_j^+},\mu),\; E_1^{\rightarrow}(x_{B_j^-},\mu) + E_2^{\leftarrow}(x_{B_j^+},\mu)\bigr\},
\end{aligned}$$

a term with an empty run being read as 0, and therefore

$$D(q, a_j + \mu) = \max\bigl\{D^{\rightarrow}(x_{B_j^-},\mu),\; D^{\leftarrow}(x_{B_j^+},\mu)\bigr\}, \qquad |\mu| < \eta. \tag{S2}$$

If $j = s$ then $C \subseteq B_s$, a single run, and $C \in \mathcal{C}_k(K)$ iff $C \in \mathcal{C}_k(|B_s|)$ after identification, so $D(q, a_s + \mu) = D(x_{B_s}, \mu)$ for $|\mu| < \eta$.

Step 3. For any run R and $|\mu| \geq \eta > \|x_R\|_\infty$, all $x_i - \mu$, $i \in R$, have the same sign, the maximum of $x_R(C) - |C|\mu$ over all $C \subseteq R$ is attained at $\emptyset$ or at R, both in the first family, and the corresponding D-function vanishes. Hence $\Phi_R(x_R) = \sup_{|\mu|<\eta} D^{(R)}(x_R, \mu)$, where $D^{(R)}$ is D, $D^{\rightarrow}$ or $D^{\leftarrow}$ according to the type of R.

Combining the three steps, $\lambda \mapsto D(q,\lambda)$ vanishes outside the disjoint intervals $(a_j - \eta, a_j + \eta)$, and on the jth of these it is the maximum of the D-functions of the runs at level a_j; taking the supremum over λ gives $\Delta(q) = \max_R \Phi_R(x_R)$.

The listed properties of Δ_p' follow from Lemma S1: a maximum of finitely many functions that are continuous, positively homogeneous, invariant under constant shifts, 2-Lipschitz for $\|\cdot\|_1$ and piecewise linear on finite fans of polyhedral cones has the same properties (the common refinement of the fans is a finite fan). The runs and their types are functions of $\sigma(p)$, so is Δ_p'. By Lemma S2, $\Phi_R \not\equiv 0$ if and only if $|R| \geq 2$ and R is not a top run of exactly two cells, which is exactly the condition that R be the vertex set of a component of $\mathcal{T}(p)$ with at least one edge; and $\Delta_p' \equiv 0$ iff every $\Phi_R \equiv 0$ iff $\mathcal{T}(p) = \emptyset$. □

Proof of (a). Take $r = \eta(p)$, so that $\|x\| < r$ implies $\|x\|_\infty < \eta(p)$ for the Euclidean norm,

and Δ'_p as in (S1). Lemma S3 gives $\Delta(p+x) = \Delta'_p(x)$ for all such x, in particular for those with $p + x \in \mathcal{S}$, together with continuity, positive homogeneity and $\Delta'_p \equiv 0$ when $\mathcal{T}(p) = \emptyset$. The lemma also shows what Δ'_p depends on: the components of $\mathcal{T}(p)$ and, for each component, whether it is a left run, the top run or a right run of p. The last piece of information is not a function of $\mathcal{T}(p)$ alone ; it is a function of $\sigma(p)$. Lemma S2 makes the remark following the definition of a contact precise: a flat stretch with a higher neighbour on one side lets a perturbation dig a valley as soon as it has two cells ($\Delta^{\rightarrow}(1,0) > 0$), a top plateau does so as soon as it has three ($\Delta(1,0,1) > 0$), and a top plateau of two cells never does ($\Delta \equiv 0$ on $\mathbb{R}^2$), whichever end of the cell range it touches. □

S1.3 Proof of (b)

Let $p \in \mathcal{U}$ with $\mathcal{T}(p) = \emptyset$. By the law of large numbers $\mathbb{P}(\|\hat{p} - p\|_\infty < \eta(p)) \to 1$, and on this event Lemma S3 gives $\Delta(\hat{p}) = \Delta'_p(\hat{p} - p) = 0$. Since $c_n^* \geq 0$, rejection requires $\Delta(\hat{p}) > 0$, so $\mathbb{P}(\text{reject}) \leq \mathbb{P}(\|\hat{p} - p\|_\infty \geq \eta(p)) \to 0$. □

S1.4 Proof of (c)

Lemma S4 (tie recovery)**.** *Let $p \in \mathcal{U}$, $\eta = \eta(p)$, and let A_n be the event that the maximal runs of pooled cells coincide with the runs of p. Let $B_n = A_n \cap \{\|\hat{p} - p\|_\infty < \eta/2\}$. Then $\mathbb{P}(B_n) \to 1$, and on B_n:*

(i) the pooled vector $\tilde{p}_n$ satisfies $\tilde{p}_{n,i} = |R|^{-1} \sum_{i' \in R} \hat{p}_{i'}$ for i in the run R of p, $\tilde{p}_n \in \mathcal{S}$, and $\|\tilde{p}_n - p\|_\infty \leq \|\hat{p} - p\|_\infty$;

(ii) $\sigma(\tilde{p}_n) = \sigma(p)$; hence $\tilde{p}_n \in \mathcal{U}$, $\mathcal{T}(\tilde{p}_n) = \mathcal{T}(p)$, and $\bar{p}_n = \tilde{p}_n$;

(iii) $u_n = \bar{p}_n - p$ is constant on each run of p and $\|u_n\|_\infty < \eta/2$, and consequently

$$\Delta(\bar{p}_n + x) = \Delta'_p(x) \qquad \text{for all } x \in \mathbb{R}^K \text{ with } \|x\|_\infty < \eta/2.$$

Proof. The pair $(k, k+1)$ is pooled iff $|d_k| \leq \tau_k$, where $d_k = \hat{p}_{k+1} - \hat{p}_k$ and $\tau_k = \kappa_n\{(\hat{p}_k + \hat{p}_{k+1})/n\}^{1/2}$. Suppose $p_k = p_{k+1}$. If $p_k = 0$ then $\hat{p}_k = \hat{p}_{k+1} = 0$ almost surely and the pair is pooled. If $p_k > 0$, then $\sqrt{n}\, d_k \xrightarrow{d} Z_{k+1} - Z_k$ by the central limit theorem while $\sqrt{n}\, \tau_k = \kappa_n(\hat{p}_k + \hat{p}_{k+1})^{1/2} \xrightarrow{p} \infty$, so for every M,

$$\begin{aligned}\limsup_n \mathbb{P}(|d_k| > \tau_k) &\leq \limsup_n \big[\mathbb{P}(\sqrt{n}|d_k| > M) + \mathbb{P}(\sqrt{n}\,\tau_k \leq M)\big] \\ &\leq \mathbb{P}(|Z_{k+1} - Z_k| \geq M),\end{aligned}$$

which tends to 0 as $M \to \infty$. Suppose $p_k \neq p_{k+1}$. Then $|d_k| \xrightarrow{p} |p_{k+1} - p_k| \geq \delta(p) > 0$ while $\tau_k \leq \kappa_n/\sqrt{n} \to 0$, so $\mathbb{P}(|d_k| \leq \tau_k) \to 0$. The pooled runs are the maximal runs of consecutive pooled pairs, so A_n is the intersection over $k < K$ of the events $\{(k, k+1) \text{ pooled}\} \triangle \{p_k \neq p_{k+1}\}$, each of probability tending to one; with the law of large numbers, $\mathbb{P}(B_n) \to 1$.

(i) On A_n the averaging is over the runs of p. Averaging preserves nonnegativity and the total mass, so $\tilde{p}_n \in \mathcal{S}$; since p is constant on each run, $\tilde{p}_{n,i} - p_i$ is the average of $\hat{p}_{i'} - p_{i'}$ over the run containing i, and the sup-norm bound follows.

(ii) Adjacent cells in the same run of p receive the same value. Adjacent cells $k, k+1$ in different runs satisfy $|p_{k+1} - p_k| \geq \delta(p) \geq 4K\eta > \eta$ and $|(\tilde{p}_{n,k+1} - \tilde{p}_{n,k}) - (p_{k+1} - p_k)| \leq 2\|\tilde{p}_n - p\|_\infty < \eta$, so $\tilde{p}_{n,k+1} - \tilde{p}_{n,k}$ has the sign of $p_{k+1} - p_k$. Thus $\sigma(\tilde{p}_n) = \sigma(p)$. Unimodality is a property of the sign pattern, so $\tilde{p}_n \in \mathcal{U}$, and $\mathcal{T}$ is a function of the sign pattern (the runs and the top run are), so $\mathcal{T}(\tilde{p}_n) = \mathcal{T}(p)$. The least-squares projection of a point of $\mathcal{U}$ onto $\mathcal{U}$ is that point. The implemented projection computes, for each candidate mode m, the isotonic fits of $\tilde{p}_n$ on $\{1, \ldots, m-1\}$ and on $\{m+1, \ldots, K\}$, sets the value at m to the maximum of $\tilde{p}_{n,m}$ and the two adjacent fitted values, and returns the fit with the smallest residual sum of squares, clipped at zero and renormalised. For $\tilde{p}_n \in \mathcal{U}$ and m a mode of $\tilde{p}_n$, the fit is $\tilde{p}_n$ itself with residual zero; a residual of zero identifies the fit, and clipping and renormalising a vector of $\mathcal{S}$ leave it unchanged. Hence $\bar{p}_n = \tilde{p}_n$.

(iii) By (i) and (ii), $u_n = \tilde{p}_n - p$ is constant on each run of p and $\|u_n\|_\infty \leq \|\hat{p} - p\|_\infty < \eta/2$. If $\|x\|_\infty < \eta/2$ then $\|u_n + x\|_\infty < \eta$, and Lemma S3 gives $\Delta(\bar{p}_n + x) = \Delta(p + u_n + x) = \Delta'_p(u_n + x) = \max_R \Phi_R(u_{n,R} + x_R) = \max_R \Phi_R(x_R) = \Delta'_p(x)$, the fourth equality because $u_{n,R}$ is a constant vector and each Φ_R is invariant under constant shifts (Lemma S1(ii)). □

Part (iii) is the uniform-radius statement: the bootstrap centre is not merely close to p with the same contacts, it differs from p by a vector that is constant on the runs of p, and such a difference is invisible to the local function. Note also that on B_n the projection step is the identity; the only role of the projection is to guarantee $\bar{p}_n \in \mathcal{U}$ on the complement of B_n, which matters in (d).

Lemma S5 (conditional convergence). *(i) Let π_n be $\mathcal{S}$-valued functions of $\hat{p}$ with $\pi_n \xrightarrow{p} p$, and given $\hat{p}$ let $\hat{p}^*_n = N^*_n/n$ with $N^*_n \sim \text{Mult}(n, \pi_n)$. Then*

$$d_{\text{BL}}\big(\mathcal{L}^*(\sqrt{n}(\hat{p}^*_n - \pi_n)),\, N(0, \Sigma(p))\big) \xrightarrow{p} 0.$$

(ii) If $d_{\text{BL}}(\mathcal{L}^*(W^*_n), \mathcal{L}(W)) \xrightarrow{p} 0$ for random vectors W^*_n, W in $\mathbb{R}^K$, and $g : \mathbb{R}^K \to \mathbb{R}$ is continuous, then $d_{\text{BL}}(\mathcal{L}^*(g(W^*_n)), \mathcal{L}(g(W))) \xrightarrow{p} 0$.

(iii) *If* X_n^*, Y_n^* *are real random variables with* $\mathbb{P}^*(X_n^* \neq Y_n^*) \xrightarrow{p} 0$, *then* $d_{\mathrm{BL}}(\mathcal{L}^*(X_n^*), \mathcal{L}^*(Y_n^*)) \xrightarrow{p} 0$.

(iv) *If* G_n *are random laws on* $\mathbb{R}$ *and* F *is a law on* $\mathbb{R}$ *with* $d_{\mathrm{BL}}(G_n, F) \xrightarrow{p} 0$, *then* $G_n((-\infty, c]) \xrightarrow{p} F((-\infty, c])$ *at every continuity point* c *of* F.

Proof. (i) Given a subsequence, extract a further subsequence along which $\pi_n \to p$ almost surely, and fix an ω in the corresponding event. Along it, $\mathcal{L}^*(\sqrt{n}(\hat{p}_n^* - \pi_n))$ is the law of $n^{-1/2}\sum_{i\le n}(\xi_{ni} - \pi_n)$ with ξ_{ni} i.i.d. $\mathrm{Mult}(1, \pi_n)$, $\pi_n = \pi_n(\omega)$ deterministic. The summands are centred, bounded by $\sqrt{2}/\sqrt{n}$ in Euclidean norm, with covariance $\Sigma(\pi_n) \to \Sigma(p)$; the Lindeberg condition holds trivially and the Lindeberg–Feller theorem with the Cramér–Wold device gives weak convergence to $N(0, \Sigma(p))$, hence $d_{\mathrm{BL}} \to 0$. Almost sure convergence along a further subsequence of every subsequence is convergence in probability.

(ii) By the same subsequence device it suffices to note that $d_{\mathrm{BL}}(P_n, P) \to 0$ and g continuous imply $d_{\mathrm{BL}}(P_n \circ g^{-1}, P \circ g^{-1}) \to 0$, which is the continuous mapping theorem.

(iii) For $\|h\|_\infty \le 1$, $|\mathrm{E}^* h(X_n^*) - \mathrm{E}^* h(Y_n^*)| \le 2\mathbb{P}^*(X_n^* \neq Y_n^*)$.

(iv) For $\varepsilon > 0$ let $h_\varepsilon^\pm$ be Lipschitz with $\mathbf{1}_{(-\infty, c-\varepsilon]} \le h_\varepsilon^- \le \mathbf{1}_{(-\infty, c]} \le h_\varepsilon^+ \le \mathbf{1}_{(-\infty, c+\varepsilon]}$. Then $\int h_\varepsilon^\pm dG_n \xrightarrow{p} \int h_\varepsilon^\pm dF$, so with probability tending to one $F(c-\varepsilon) - \varepsilon \le G_n((-\infty, c]) \le F(c+\varepsilon) + \varepsilon$; let $\varepsilon \downarrow 0$ and use continuity of F at c. □

Lemma S6 (the limit law)**.** *Let* $p \in \mathcal{U}$ *with* $\min_i p_i > 0$ *and* $\mathcal{T}(p) \neq \emptyset$, *let* $Z \sim N(0, \Sigma(p))$, *and let* F *be the distribution function of* $\Delta_p'(Z)$. *Then:*

(i) $\Delta_p'(Z)$ *has no atom in* $(0, \infty)$, *so* F *is continuous on* $(0, \infty)$;

(ii) $F(c) > F(q)$ *whenever* $0 \le q < c$; *in particular* $\mathbb{P}(\Delta_p'(Z) > 0) > 0$;

(iii) *with* $N_0 = \{x \in H : \Delta_p'(x) = 0\}$, *closed in* H, $\mathbb{P}(Z \in \partial_H N_0) = 0$, ∂_H *denoting the boundary relative to* H.

Proof. For $v \in \mathbb{R}^K$, $v^\top \Sigma(p) v = \sum_i p_i v_i^2 - (\sum_i p_i v_i)^2$ is the variance of v_J for $J \sim \mathrm{Mult}(1, p)$, which vanishes iff v is constant on $\{i : p_i > 0\} = [K]$. So the null space of $\Sigma(p)$ is $\mathbb{R}\mathbf{1}$, $Z \in H$ almost surely, and Z has an everywhere positive density with respect to Lebesgue measure on H.

(i) Let $Q_1, \dots, Q_M$ and $g_1, \dots, g_M$ be as in Lemma S1(iv) for Δ_p' (Lemma S3). If $Q_j \cap H$ has dimension less than $K-1$ it is Z-null. Otherwise, on $\{Z \in Q_j\}$, $\Delta_p'(Z) = g_j^\top Z$, and $g_j^\top Z$ is either degenerate at 0 (if $g_j \in \mathbb{R}\mathbf{1}$) or normal with positive variance; in both cases $\mathbb{P}(g_j^\top Z \in N) = 0$ for every Lebesgue-null $N \subseteq (0, \infty)$. Summing over j, $\mathbb{P}(\Delta_p'(Z) \in N) = 0$.

(ii) By Lemma S3 there is x_0 with $\Delta_p'(x_0) > 0$; by shift invariance and homogeneity we may take $x_0 \in H$ and $\Delta_p'(x_0) = 1$. The set $\{x \in H : q < \Delta_p'(x) < c\}$ is open in H and contains tx_0 for $q < t < c$, so it has positive Z-probability.

(iii) N_0 is the intersection with H of the finite union of the closed polyhedral cones $Q_j \cap \{g_j^\top x = 0\}$. The boundary relative to H of a finite union is contained in the union of the boundaries, and the boundary relative to H of a polyhedral cone intersected with H is contained in finitely many hyperplanes of H (in the cone itself, contained in a hyperplane of H, when it is not of full dimension in H). Hyperplanes of H are Z-null. □

Proof of (c). Let $p \in \mathcal{U}$ with $\mathcal{T}(p) \neq \emptyset$ and $\min_i p_i > 0$, $\eta = \eta(p)$, $W_n = \sqrt{n}(\hat{p} - p)$ and $T_n = \sqrt{n}\,\Delta(\hat{p})$. By the multinomial central limit theorem $W_n \xrightarrow{d} Z$, and W_n, Z take values in H.

The statistic. On $\{\|\hat{p} - p\|_\infty < \eta\}$, whose probability tends to one, Lemma S3 and homogeneity give $T_n = \sqrt{n}\,\Delta_p'(\hat{p}-p) = \Delta_p'(W_n)$. Since Δ_p' is continuous, $\Delta_p'(W_n) \xrightarrow{d} \Delta_p'(Z)$, and $\mathbb{P}(T_n \neq \Delta_p'(W_n)) \to 0$, so $T_n \xrightarrow{d} \Delta_p'(Z)$.

The bootstrap. Let $W_n^* = \sqrt{n}(\hat{p}^* - \bar{p}_n)$ and $X_n^* = \sqrt{n}\,\Delta(\hat{p}^*)$. On B_n of Lemma S4, $\|\bar{p}_n - p\|_\infty \le \|\hat{p} - p\|_\infty$, so $\bar{p}_n \xrightarrow{p} p$, and Lemma S5(i) with $\pi_n = \bar{p}_n$ gives $d_{\mathrm{BL}}(\mathcal{L}^*(W_n^*), \mathcal{L}(Z)) \xrightarrow{p} 0$. On $B_n \cap \{\|\hat{p}^* - \bar{p}_n\|_\infty < \eta/2\}$, Lemma S4(iii) and homogeneity give $X_n^* = \Delta_p'(W_n^*)$. By Chebyshev's inequality, $\mathbb{P}^*(\|\hat{p}^* - \bar{p}_n\|_\infty \ge \eta/2) \le \sum_j \mathrm{Var}^*(\hat{p}_j^*)/(\eta/2)^2 \le 4K/(n\eta^2)$, so $\mathbb{P}^*(X_n^* \neq \Delta_p'(W_n^*)) \le \mathbf{1}_{B_n^c} + 4K/(n\eta^2) \xrightarrow{p} 0$. Lemma S5(ii),(iii) give $d_{\mathrm{BL}}(\mathcal{L}^*(X_n^*), \mathcal{L}(\Delta_p'(Z))) \xrightarrow{p} 0$, which is the second assertion of (c).

Size. Let F be the distribution function of $\Delta_p'(Z)$, $q = \inf\{t : F(t) \ge 1 - \alpha\}$ its $(1-\alpha)$ quantile, $G_n = \mathcal{L}^*(X_n^*)$ and $c_n = \inf\{t : G_n((-\infty, t]) \ge 1 - \alpha\} = \sqrt{n}\,c_n^*$. The test rejects iff $T_n > c_n$. By Lemma S6(i), every $c > 0$ is a continuity point of F; by Lemma S5(iv), $G_n((-\infty, c]) \xrightarrow{p} F(c)$ for every $c > 0$. If $F(c) < 1 - \alpha$ then $G_n((-\infty, c]) < 1 - \alpha$ with probability tending to one, and on that event $c_n \ge c$; if $F(c) > 1 - \alpha$ then likewise $c_n \le c$ with probability tending to one.

Case $q > 0$. F is continuous at q, so $F(q) = 1 - \alpha$. For $0 < c < q$, $F(c) < 1 - \alpha$ by definition of q; for $c > q$, $F(c) > F(q) = 1 - \alpha$ by Lemma S6(ii). Hence $c_n \xrightarrow{p} q$, so $T_n - c_n \xrightarrow{d} \Delta_p'(Z) - q$, and since $\mathbb{P}(\Delta_p'(Z) = q) = 0$,

$$\mathbb{P}(\text{reject}) = \mathbb{P}(T_n - c_n > 0) \to \mathbb{P}(\Delta_p'(Z) > q) = 1 - F(q) = \alpha.$$

Case $q = 0$, that is $F(0) = \mathbb{P}(\Delta_p'(Z) = 0) \ge 1 - \alpha$. Since $c_n \ge 0$, $\mathbb{P}(\text{reject}) \le \mathbb{P}(T_n > 0)$. Now $\mathbb{P}(T_n = 0) = \mathbb{P}(W_n \in N_0) + o(1)$ with N_0 as in Lemma S6(iii), and $W_n \xrightarrow{d} Z$ as random elements of H with $\mathbb{P}(Z \in \partial_H N_0) = 0$, so by the portmanteau theorem $\mathbb{P}(T_n = 0) \to \mathbb{P}(Z \in N_0) = F(0)$. Thus $\limsup_n \mathbb{P}(\text{reject}) \le 1 - F(0) \le \alpha$. Conversely, for every

$c > 0$, $F(c) > F(0) \geq 1 - \alpha$ by Lemma S6(ii), so $c_n \leq c$ with probability tending to one and $\liminf_n \mathbb{P}(\text{reject}) \geq \lim_n \mathbb{P}(T_n > c) = 1 - F(c)$; letting $c \downarrow 0$ and using right-continuity, $\mathbb{P}(\text{reject}) \to 1 - F(0)$.

In both cases $\limsup_n \mathbb{P}(\text{reject}) \leq \alpha$. The law of $\Delta'_p(Z)$ is continuous at q exactly when $q > 0$ (when $q = 0$ it has an atom of mass $F(0) \geq 1 - \alpha > 0$ at q), and then the limit equals α. Altogether,

$$\lim_n \mathbb{P}(\text{reject}) = \min\bigl\{\alpha,\ \mathbb{P}(\Delta'_p(Z) > 0)\bigr\},$$

which is strictly less than α precisely when the atom at zero exceeds $1 - \alpha$. □

S1.5 Proof of (d)

Let $p \notin \mathcal{U}$. By Lemma S1(iii) Δ is continuous on $\mathcal{S}$, so $\Delta(\hat{p}) \to \Delta(p)$ almost surely by the strong law of large numbers, and $\Delta(p) > 0$ by Lemma S2(i). The centre $\bar{p}_n$ lies in $\mathcal{U} \cap \mathcal{S}$ by construction (the projection returns isotonic fits on either side of a mode, clipped at zero and renormalised, which is a unimodal vector of $\mathcal{S}$), so $\Delta(\bar{p}_n) = 0$ and, by Lemma S1(iii),

$$\Delta(\hat{p}^*) \leq 2\|\hat{p}^* - \bar{p}_n\|_1, \qquad \mathrm{E}^*\|\hat{p}^* - \bar{p}_n\|_1 \leq \sum_{j=1}^{K} (\bar{p}_{n,j}/n)^{1/2} \leq (K/n)^{1/2},$$

the middle step because $\mathrm{E}^*|\hat{p}^*_j - \bar{p}_{n,j}| \leq (\mathrm{Var}^*\hat{p}^*_j)^{1/2}$, the last by the Cauchy–Schwarz inequality. With $b_n = 2(K/n)^{1/2}/\alpha$, Markov's inequality gives $\mathbb{P}^*(\Delta(\hat{p}^*) > b_n) \leq \alpha$, hence $c^*_n \leq b_n$, a deterministic bound tending to zero. Therefore $\mathbb{P}(\text{reject}) \geq \mathbb{P}(\Delta(\hat{p}) > b_n) \to 1$. □

S1.6 Remarks

Remark 1 (Monte Carlo approximation). The implemented test draws $\Delta^*_1, \ldots, \Delta^*_B$ independently from $\mathcal{L}^*(\Delta(\hat{p}^*))$ and rejects when $(1 + \#\{b : \Delta^*_b \geq \Delta(\hat{p})\})/(B + 1) \leq \alpha$. Parts (b) and (d) hold verbatim for every fixed B with $\alpha(B + 1) \geq 1$: under (b), $\Delta(\hat{p}) = 0$ with probability tending to one and then every $\Delta^*_b \geq \Delta(\hat{p})$, so the Monte Carlo p-value equals one; under (d), $\mathbb{P}^*(\Delta^*_b \geq \Delta(\hat{p})) \leq b_n\alpha/\Delta(\hat{p}) \xrightarrow{p} 0$ for each b, so the p-value tends to $1/(B + 1) \leq \alpha$ in probability. Under (c), the test rejects only when $\Delta(\hat{p}) > 0$, and the arguments above show that $(T_n, \sqrt{n}\Delta^*_1, \ldots, \sqrt{n}\Delta^*_B)$ converges in law to $(Y_0, \ldots, Y_B)$ i.i.d. with distribution function F, with $\mathbb{P}(T_n = 0) \to F(0)$; the limiting rejection probability is $\mathbb{P}(Y_0 > 0,\ \#\{b \geq 1 : Y_b \geq Y_0\} \leq \alpha(B + 1) - 1)$. For exchangeable $Y_0, \ldots, Y_B$ at most $m + 1$ indices i satisfy $\#\{b \neq i : Y_b \geq Y_i\} \leq m$, so this probability is at most $(\lfloor \alpha(B + 1) \rfloor)/(B + 1) \leq \alpha$.

Remark 2 (On the statement of the theorem). The local function Δ'_p depends on p through the sign pattern $\sigma(p)$ and not through the set $\mathcal{T}(p)$ alone. Let $p = (1,2,2,2,3)/10$ and $p' = (1,3,3,3,2)/12$; both have $\mathcal{T} = \{(2,3),(3,4)\}$. For $x = \varepsilon(0,-\frac{1}{2},1,-\frac{1}{2},0)$ with $\varepsilon > 0$ small, $p+x$ has the valley $p_3+\varepsilon > p_4-\varepsilon/2 < p_5$, so that $\Delta'_p(x) = \Delta^{\rightarrow}(-\frac{1}{2},1,-\frac{1}{2})\varepsilon = 3\varepsilon/4 > 0$, whereas $p'+x$ is unimodal and $\Delta'_{p'}(x) = 0$. What matters is whether each flat stretch is the top plateau or is open to a higher neighbour on one side, and this is determined by $\sigma(p)$. The condition $\min_i p_i > 0$ in (c) is used only to give Z a positive density on H, which excludes atoms of $\Delta'_p(Z)$ in $(0,\infty)$; under the null it holds whenever the first and last retained cells carry positive probability, since a unimodal vector vanishing at an interior cell vanishes on a whole end segment. Finally, the supremum over λ in the definition of Δ is a maximum over the finite set $M(p)$ of Lemma S1(i), which contains, and is in general larger than, the set of cell probabilities.

S2 Proofs for the reporting mechanism

Throughout, $\mathcal{L} = \{\ell_1 < \cdots < \ell_J\}$ with $\ell_1 = 1$, $W = \operatorname{lcm}\mathcal{L}$, and for $\ell \in \mathcal{L}$ we write $m = m_\ell = W/\ell$. For $d \mid W$ the pattern is $S(d) = \{\ell \in \mathcal{L} : \ell \mid d\}$ and $D_S = \{d \mid W : S(d) = S\}$. We use repeatedly that for $k \in \mathbb{Z}$, $\gcd(k,m) = \gcd(k \bmod m, m)$, and that for $e \mid m$ the number of residues $r \in \{0,\dots,m-1\}$ with $\gcd(r,m) = e$ is $\varphi(m/e)$ (the residue $r = 0$ counts for $e = m$).

S2.1 Proof of Lemma 1

Let f_ℓ be the conditional density of X given $L = \ell$, $\phi_\ell(s) = \int e^{isx} f_\ell(x)\,dx$, and $f_\ell^W(u) = \sum_{j\in\mathbb{Z}} f_\ell(u+jW)$, $u \in [0,W)$, its W-periodization, assumed in $L^2[0,W)$. Identify $[0,W)$ with the circle $\mathbb{T} = \mathbb{R}/W\mathbb{Z}$.

Fix a residue $r \in \{0,\dots,m-1\}$. Since $K_\ell = \lfloor x/\ell + \frac{1}{2} \rfloor$, the event $K_\ell \equiv r \pmod m$ is

$$\bigcup_{j\in\mathbb{Z}} \{x :\ r + jm \le x/\ell + \tfrac{1}{2} < r + jm + 1\} = \bigcup_{j\in\mathbb{Z}} \big[\ell r - \tfrac{\ell}{2} + jW,\ \ell r + \tfrac{\ell}{2} + jW\big) = I_r + W\mathbb{Z},$$

where $I_r = [\ell r - \ell/2, \ell r + \ell/2)$ is an arc of length ℓ on $\mathbb{T}$ (for $r = 0$ it wraps around 0). Hence, by Tonelli,

$$\mathbb{P}(K_\ell \equiv r \mid L = \ell) = \int_{\mathbb{T}} f_\ell^W(u)\, \mathbf{1}_{I_r}(u)\,du.$$

Both factors lie in $L^2(\mathbb{T})$. With $c_t(F) = W^{-1}\int_0^W F(u)e^{-2\pi i t u/W}\,du$, Tonelli gives $c_t(f_\ell^W) = W^{-1}\int_{\mathbb{R}} f_\ell(x)e^{-2\pi i t x/W}\,dx = W^{-1}\phi_\ell(-2\pi t/W)$, while a direct integration over the arc gives

$c_0(\mathbf{1}_{I_r}) = \ell/W = 1/m$ and, for $t \neq 0$,

$$c_t(\mathbf{1}_{I_r}) = \frac{1}{W}\int_{\ell r-\ell/2}^{\ell r+\ell/2} e^{-2\pi i t u/W}\,du = e^{-2\pi i t \ell r/W}\,\frac{\sin(\pi t\ell/W)}{\pi t} = e^{-2\pi i t r/m}\,\frac{\sin(\pi t/m)}{\pi t}.$$

Parseval's identity on $\mathbb{T}$, $\int_{\mathbb{T}} F\bar{G} = W\sum_{t\in\mathbb{Z}} c_t(F)\overline{c_t(G)}$, the series converging absolutely by Cauchy–Schwarz, yields

$$\mathbb{P}(K_\ell \equiv r \mid L=\ell) - \frac{1}{m} = \sum_{t\neq 0} \phi_\ell(-2\pi t/W)\, e^{2\pi i t r/m}\,\frac{\sin(\pi t/m)}{\pi t}.$$

Taking absolute values, using $|\phi_\ell(-s)| = |\phi_\ell(s)|$ and pairing t with $-t$,

$$\Big|\mathbb{P}(K_\ell \equiv r \mid L=\ell) - \frac{1}{m}\Big| \le 2\sum_{t\ge 1} \frac{|\sin(\pi t/m)|}{\pi t}\,\big|\phi_\ell(2\pi t/W)\big|. \tag{S3}$$

Since $|\sin(\pi t/m)| \le \min(\pi t/m, 1)$, the weight in (S3) is at most $\min(1/m, 1/(\pi t))$, which is the bound of Lemma 1. The factor 2 is the pairing of $\pm t$, and $1/m = \ell/W$ is the zero-frequency coefficient of the arc. □

Two remarks. The bound (S3) is slightly sharper than that of Lemma 1: its weight vanishes at $t \equiv 0 \pmod m$, that is at the harmonics $2\pi t'/\ell$ of the level itself, which cannot affect a residue modulo m of the multiplier. Second, the hypothesis $f_\ell^W \in L^2$ is used only to justify Parseval; the right-hand side is then automatically finite, since $\sum_t |\phi_\ell(2\pi t/W)|^2 < \infty$ and $\sum_t (\pi t)^{-2} < \infty$.

S2.2 Proof of the error bound in Proposition 1

Let $\varepsilon_\ell = \max_r |\mathbb{P}(K_\ell \equiv r \mid L=\ell) - 1/m|$, with $1/m = \ell/W$ as in the statement. On $\{L=\ell\}$, $G = \gcd(\ell K_\ell, \ell m) = \ell\gcd(K_\ell, m) = \ell\gcd(K_\ell \bmod m,\, m)$, so $G = d$ requires $\ell \mid d$ and then occurs exactly when $K_\ell \bmod m$ lies in the set of $\varphi(m/e) = \varphi(W/d)$ residues r with $\gcd(r,m) = e = d/\ell$. Therefore, for $\ell \mid d$,

$$\big|\mathbb{P}(G=d \mid L=\ell) - \Lambda_{d\ell}\big| = \Big|\sum_{r:\gcd(r,m)=d/\ell} \Big(\mathbb{P}(K_\ell \equiv r \mid L=\ell) - \frac{1}{m}\Big)\Big| \le \varphi(W/d)\,\varepsilon_\ell,$$

and for $\ell \nmid d$ both $\mathbb{P}(G=d \mid L=\ell)$ and $\Lambda_{d\ell}$ vanish. Averaging over L,

$$\big|\mathbb{P}(G=d) - (\Lambda\bar{\pi})_d\big| \le \sum_{\ell\in\mathcal{L}} \bar{\pi}_\ell \big|\mathbb{P}(G=d \mid L=\ell) - \Lambda_{d\ell}\big| \le \varphi(W/d)\sum_{\ell\in S(d)} \bar{\pi}_\ell\varepsilon_\ell \le \varphi(W/d)\sum_{\ell\in\mathcal{L}} \bar{\pi}_\ell\varepsilon_\ell,$$

which is the stated bound; the middle expression, with the sum restricted to $S(d)$, is slightly sharper. By Lemma 1, $\varepsilon_\ell \le 2\sum_{t\ge1}\min(1/m_\ell, 1/(\pi t))|\phi_\ell(2\pi t/W)|$. □

S2.3 Proof of Proposition 2

Let $\bar{\mathcal{L}} = \{\operatorname{lcm} T : \emptyset \ne T \subseteq \mathcal{L}\}$ be the closure of $\mathcal{L}$ under least common multiples; $\mathcal{L} \subseteq \bar{\mathcal{L}}$ and every element of $\bar{\mathcal{L}}$ divides W.

Lemma S7. *The attainable patterns are exactly the sets $S(e)$, $e \in \bar{\mathcal{L}}$, and $e \mapsto S(e)$ is a bijection from $\bar{\mathcal{L}}$ onto them. Hence $V = |\bar{\mathcal{L}}| \ge J$, with $V = J$ if and only if $\mathcal{L}$ is closed under* lcm*; in particular $V = J$ when $\mathcal{L}$ is totally ordered by divisibility.*

Proof. For $d \mid W$ put $\bar{d} = \operatorname{lcm} S(d) \in \bar{\mathcal{L}}$. Every $\ell \in S(d)$ divides d, so $\bar{d} \mid d$, whence $S(\bar{d}) \subseteq S(d)$; conversely $\ell \in S(d)$ implies $\ell \mid \bar{d}$. Thus $S(d) = S(\bar{d})$, and every attainable pattern is of the form $S(e)$, $e \in \bar{\mathcal{L}}$; every such set is attainable because $e \mid W$. If $e = \operatorname{lcm} T \in \bar{\mathcal{L}}$ then $T \subseteq S(e)$, so $e \mid \operatorname{lcm} S(e)$, and $\operatorname{lcm} S(e) \mid e$ because each element of $S(e)$ divides e; hence $\operatorname{lcm} S(e) = e$, and $S(e)$ determines e. This proves the bijection and $V = |\bar{\mathcal{L}}|$. Since $\mathcal{L} \subseteq \bar{\mathcal{L}}$, $V \ge J$, with equality if and only if $\bar{\mathcal{L}} = \mathcal{L}$, that is, if and only if $\operatorname{lcm}(\ell, \ell') \in \mathcal{L}$ for all $\ell, \ell' \in \mathcal{L}$ (pairwise closure gives closure under finite lcm by induction). A chain is closed, since then $\operatorname{lcm}(\ell, \ell') = \max(\ell, \ell')$. □

The first sentence of the proposition is the first part of the lemma, since $S(\operatorname{lcm} T)$ with T ranging over subsets of $\mathcal{L}$ ranges over $\{S(e) : e \in \bar{\mathcal{L}}\}$ (the empty T gives $S(1) = \{1\} = S(\ell_1)$).

Restrictions. Write $p_S = \mathbb{P}(S(G) = S)$ for an attainable S. Summing (1) over $d \in D_S$,

$$p_S = c_S \sum_{\ell \in S} \ell\, \bar{\pi}_\ell, \qquad c_S = \frac{1}{W} \sum_{d \in D_S} \varphi(W/d) > 0, \tag{S4}$$

so $p = B\bar{\pi}$ with B the $V \times J$ matrix $B_{S\ell} = c_S\, \ell\, \mathbf{1}\{\ell \in S\}$. Its rows indexed by $S(\ell')$, $\ell' \in \mathcal{L}$, form a $J \times J$ block with $B_{S(\ell')\ell} \ne 0$ if and only if $\ell \mid \ell'$; ordering $\mathcal{L}$ increasingly, the block is lower triangular with positive diagonal, so $\operatorname{rank} B = J$. The columns of B sum to one: $\sum_S B_{S\ell} = W^{-1}\ell \sum_{d:\ \ell \mid d \mid W} \varphi(W/d) = W^{-1}\ell \sum_{e \mid m} \varphi(m/e) = \ell m/W = 1$. Consequently B maps the hyperplane $\{\mathbf{1}^\top \pi = 1\}$ of $\mathbb{R}^J$ injectively into the hyperplane $\{\mathbf{1}^\top p = 1\}$ of $\mathbb{R}^V$, and the set of pattern distributions compatible with the model is the intersection of the simplex Δ_V with the $(J-1)$-dimensional affine subspace $\mathcal{A} = B\{\mathbf{1}^\top \pi = 1\}$, further cut by the J inequalities $B^- p \ge 0$, B^- any left inverse of B. Within the $(V-1)$-dimensional affine hull of Δ_V, $\mathcal{A}$ has codimension $V - J$: the model imposes exactly $V - J$ linear equality restrictions on the pattern distribution, and beyond them only the inequalities $\bar{\pi} \ge 0$.

The case $V = J$. Then B is square and invertible, $\mathcal{A}$ is the whole hyperplane, and there are no equality restrictions. Let N_S be the pattern counts and $\hat{p} = (N_S/n)_S$. By Proposition 1(b) the likelihood of the counts of G is the multinomial likelihood of the pattern counts times a factor free of $\bar{\pi}$, so the two maximum-likelihood fits coincide. The multinomial log-likelihood $\sum_S N_S \log p_S$ is maximized over Δ_V uniquely at $\hat{p}$. Put $\hat{\pi} = B^{-1}\hat{p}$; since $\mathbf{1}^\top B = \mathbf{1}^\top$, also $\mathbf{1}^\top B^{-1} = \mathbf{1}^\top$ and $\mathbf{1}^\top \hat{\pi} = 1$. If $\hat{\pi} \geq 0$, then $\hat{\pi} \in \Delta_J$ and $B\hat{\pi} = \hat{p}$ attains the unconstrained maximum, so $\hat{\pi}$ is the maximum-likelihood estimate and the fitted pattern frequencies $B\hat{\pi}$ equal the observed ones. □

Numerical values. Using Lemma S7 one computes: for $\{1, 100, 1000, 5000, 10^4\}$, a chain, $V = J = 5$; for the fifteen-level ladder $\{1, 2, 5, 10, 25, 50, 100, 250, 500, 1000, 2500, 5000, 10^4, 2.5{\cdot}10^4, 5{\cdot}10^4\}$, which is not a chain but is closed under lcm ($\mathrm{lcm}(2,5) = 10$, $\mathrm{lcm}(2,25) = 50$, $\mathrm{lcm}(100,250) = 500$, $\mathrm{lcm}(1000,2500) = 5000$, $\mathrm{lcm}(10^4, 2.5 \cdot 10^4) = 5 \cdot 10^4$, and the remaining pairs are comparable), $V = J = 15$; for $\{1, 10, 100, 500, 1000, 2500, 5000, 10^4, 2.5 \cdot 10^4\}$, $\bar{\mathcal{L}}$ adds the single element $\mathrm{lcm}(10^4, 2.5 \cdot 10^4) = 5 \cdot 10^4$, so $V = 10 = J + 1$, the extra pattern being the full set $S(5 \cdot 10^4) = \mathcal{L}$.

S2.4 Proof of Theorem 1

The residue law. On $\{L = \ell\}$, $Y = \ell K_\ell$ and $R = Y \bmod W = \ell\,(K_\ell \bmod m)$, because $0 \leq \ell(K_\ell \bmod m) < \ell m = W$ and $Y - \ell(K_\ell \bmod m) = W\lfloor K_\ell/m \rfloor \in W\mathbb{Z}$. Under Assumption 1, R given $L = \ell$ is therefore uniform on the m multiples of ℓ in $\{0, \ldots, W-1\}$, and

$$\mathbb{P}(R = r) = \sum_{\ell \in \mathcal{L}} \bar{\pi}_\ell \, \frac{\mathbf{1}\{\ell \mid r\}}{m_\ell} = \frac{1}{W} \sum_{\ell \in S(r)} \ell \, \bar{\pi}_\ell .$$

Since every $\ell \in \mathcal{L}$ divides W, $\ell \mid Y$ if and only if $\ell \mid R$, so $S(R) = S(Y) = S(G)$, and the right-hand side depends on r only through $S(r)$. For an attainable S with $\mathbb{P}(S(R) = S) = |C_S| W^{-1} \sum_{\ell \in S} \ell \bar{\pi}_\ell > 0$, dividing gives (3). Grouping C_S by $\gcd(r, W)$ recovers (2): the number of $r \in C_S$ with $\gcd(r, W) = d$ is $\varphi(W/d)$ for $d \in D_S$, and $|C_S| = \sum_{d \in D_S} \varphi(W/d)$.

(a) Conditional structure. Let $S_i = S(R_i)$ and let $J_i \in \{1, \ldots, M\}$ be the index of the arc containing R_i. The pairs (S_i, J_i) are independent and identically distributed, and by (3), for every S with $\rho_S = \mathbb{P}(S(R) = S) > 0$,

$$\mathbb{P}(J_i = j \mid S_i = S) = \sum_{r \in C_S \cap \mathrm{arc}_j} \frac{1}{|C_S|} = p_{j|S},$$

a quantity determined by $\mathcal{L}$, W and M alone. Hence, conditionally on $(S_1, \ldots, S_n)$, the J_i are independent with $J_i \sim p_{\cdot|S_i}$, and the array (N_{Sj}) has conditional law $\bigotimes_S \mathrm{Mult}(N_S, p_{\cdot|S})$.

This law is a function of $(N_S)_S$ only, so by the tower property it is also the conditional law given $(N_S)_S$. It does not involve $\bar{\pi}$ or $\pi_\ell(\cdot)$, which enter only the law of $(N_S)_S$.

The same holds for any rule that retains a set $\mathcal{S}_n$ of classes measurable with respect to $(N_S)_S$, such as $\mathcal{S}_n = \{S : N_S \geq 20\}$: conditionally on $(N_S)_S$ the retained set is fixed and $T_n = \sum_{S \in \mathcal{S}_n} T_{n,S}$, where $T_{n,S} = \sum_{j: p_{j|S} > 0} (N_{Sj} - N_S p_{j|S})^2 / (N_S p_{j|S})$ are conditionally independent Pearson statistics of multinomials with known parameters. (Arcs with $p_{j|S} = 0$ carry no reports under the null, and omitting them from the inner sum loses nothing; a rule that depended on the N_{Sj} themselves would not preserve the conditional law.) The conditional distribution of T_n is therefore known exactly, and the conditional Monte Carlo p-value $\hat{p} = (1 + \#\{b \leq B : T^{(b)} \geq T_n\})/(B+1)$, with $T^{(1)}, \ldots, T^{(B)}$ drawn from it, satisfies $\mathbb{P}(\hat{p} \leq \alpha \mid (N_S)_S) \leq \alpha$ for every n, since $T_n, T^{(1)}, \ldots, T^{(B)}$ are conditionally exchangeable.

(a) Asymptotics. By the strong law, $N_S/n \to \rho_S$ almost surely, so $N_S \to \infty$ and $S \in \mathcal{S}_n$ eventually when $\rho_S > 0$, while $N_S = 0$ almost surely when $\rho_S = 0$. For a class with $\rho_S > 0$, Pearson's theorem applied along the deterministic sequence $N_S \to \infty$ shows that the law of $T_{n,S}$ under $\mathrm{Mult}(N_S, p_{\cdot|S})$ converges to $\chi^2_{M_S - 1}$, $M_S = \#\{j : p_{j|S} > 0\}$. By conditional independence, the conditional characteristic function of T_n given $(N_S)_S$ is the product over $S \in \mathcal{S}_n$ of those of the $T_{n,S}$, and converges almost surely to that of χ^2_ν,

$$\nu = \sum_{S:\ \rho_S > 0} (M_S - 1),$$

which is the sum $\sum_S (M_S - 1)$ of the theorem, understood over the classes that occur; classes with $\rho_S = 0$ never contribute. The unconditional characteristic function converges by dominated convergence, so $T_n \xrightarrow{d} \chi^2_\nu$, and the conditional $(1-\alpha)$-quantiles of T_n converge almost surely to $\chi^2_{\nu, 1-\alpha}$. This proves (a).

(b) Consistency. Let $\tilde{p}_{j|S} = \mathbb{P}(J_i = j \mid S_i = S)$ under the true law of Y, and suppose $\tilde{p}_{\cdot|S} \neq p_{\cdot|S}$ for some S with $\rho_S > 0$. There is an arc j with $p_{j|S} > 0$ and $\tilde{p}_{j|S} \neq p_{j|S}$: otherwise $\tilde{p}$ and p agree on $\{j : p_{j|S} > 0\}$, where p has total mass one, forcing $\tilde{p} = p$. By the strong law $N_{Sj}/n \to \rho_S \tilde{p}_{j|S}$ and $N_S/n \to \rho_S$ almost surely, hence

$$\frac{T_n}{n} \ \to\ \kappa = \sum_{S: \rho_S > 0} \rho_S \sum_{j:\ p_{j|S} > 0} \frac{(\tilde{p}_{j|S} - p_{j|S})^2}{p_{j|S}} \quad \text{a.s.},$$

and $\kappa > 0$ by the choice of (S, j); classes with $\rho_S > 0$ are eventually retained and the others contribute nothing. Whatever the alternative, the conditional null law of T_n given $(N_S)_S$ is still $\sum_{S \in \mathcal{S}_n}$ of independent Pearson statistics with $N_S \to \infty$ for $\rho_S > 0$, so its quantiles converge almost surely to those of a fixed chi-square law. Since $T_n \to \infty$ almost surely, the

rejection probability tends to one, whether the critical value is the chi-square quantile or the conditional Monte Carlo quantile; with B draws, $\hat{p} \to 1/(B+1)$ almost surely, which is below α once $B \geq \alpha^{-1} - 1$. □

S3 Details for Lemma 2

We record the two facts that the proof in Section 3 uses. First, if w/ℓ is odd, the cell boundaries $b = kw \pm w/2 = \ell\big(kw/\ell \pm \frac{1}{2}(w/\ell)\big)$ are of the form $\ell(j + \frac{1}{2})$, $j \in \mathbb{Z}$, which are the boundaries of the rounding intervals $[\ell(j - \frac{1}{2}), \ell(j + \frac{1}{2}))$ of level ℓ; each cell is a union of complete rounding intervals, and the report ℓj lies in the same interval as the latent amount. If w/ℓ is even, both boundaries of every cell are multiples of ℓ, the interval $[b - \ell/2, b + \ell/2)$ is the rounding interval of b, and no other rounding interval meets a boundary. Second, the bound $|h_\ell(b+t) - h_\ell(b-t)| \leq 2t \sup |h'_\ell|$ requires h_ℓ absolutely continuous with h'_ℓ essentially bounded on $[b-\ell/2, b+\ell/2]$; the suprema in the lemma are essential suprema of the almost-everywhere derivative, and the bound is void, not false, when they are infinite. The lemma also uses that $\pi_\ell \equiv 0$ for $\ell \notin A$, which is the definition of A as the set of levels in use. For the cell to the left of b the splitting convention adds $\frac{1}{2}\int_b^{b+\ell/2} h_\ell$ and removes $\frac{1}{2}\int_{b-\ell/2}^{b} h_\ell$, giving the displayed error $\frac{1}{2}\int_0^{\ell/2}\{h_\ell(b+t) - h_\ell(b-t)\}\,dt$, and $\frac{1}{2}\int_0^{\ell/2} 2t\,dt = \ell^2/8$.

S4 Which profile the inversion needs

The map (5) involves $\pi_\ell(c) = \mathbb{P}(L = \ell \mid X \in C_c)$, the profile conditional on latent income. An estimator built by conditioning on the report returns a different function.

Proposition S1. *(a) $\mathbb{P}(L = \ell \mid Y = y) = 0$ whenever $\ell \nmid y$. Hence $\mathbb{P}(L = \cdot \mid Y = y)$ and $\mathbb{P}(L = \cdot \mid X = y)$ differ at every y not divisible by every level at which some $\pi_\ell(y)$ with $\ell \nmid y$ is positive, and the difference does not depend on the sample size.*

(b) Let $\mathcal{L}$ be closed under least common multiples, let $B = [a, a+h)$ with a and h multiples of W, and suppose Assumption 1 holds conditionally on $X \in B$. Then the mixture weights of Proposition 1 computed from the reports in B equal $\pi_\ell^B = \mathbb{P}(L = \ell \mid X \in B)$ up to an error bounded by

$$\frac{1}{\mathbb{P}(X \in B)} \sum_{\ell \in \mathcal{L}} \ell \sup_{x \in \partial B \pm \ell/2} h_\ell(x),$$

the mass that rounding carries across the two edges of the band.

Part (a) is immediate, since a report is a multiple of the level that produced it, and its consequence is easiest to see at a number. Under the mechanism of the simulations of Section 6, at \$12,000 the latent-conditional profile over the levels $\{1, 100, 1000, 5000, 10^4\}$ is $(0.153, 0.136, 0.339, 0.169, 0.203)$, while the report-conditional one is $(0, 0.038, 0.961, 0, 0)$. Regressing the EM posterior of each report on the report, as in the stochastic EM of Groß and Rendtel (2016), estimates a smoothed version of the second function. Part (b) shows that the right object is nevertheless identified: fitting the mixture separately on bands of reports whose edges are multiples of W estimates band averages of the latent-conditional profile, up to the mass that rounding carries across the band edges and to the variation of the profile within a band. The band must span several periods of W for the edge term to be small, which is the same requirement, applied locally, that Assumption 1 imposes globally.

S5 Proof of Proposition S1

(a). On $\{L = \ell\}$, $Y = \ell K_\ell \in \ell\mathbb{Z}$, so $\mathbb{P}(L = \ell, Y = y) = 0$ and $\mathbb{P}(L = \ell \mid Y = y) = 0$ whenever $\ell \nmid y$ and $\mathbb{P}(Y = y) > 0$. The latent-conditional profile at y is $\pi_\ell(y)$, which is not constrained by divisibility. The two profiles therefore differ at every y for which $\pi_\ell(y) > 0$ for some $\ell \nmid y$; in particular, if every $\pi_\ell(\cdot)$ is positive, at every y not divisible by every level. Both are population quantities, so the difference is unaffected by the sample size. □

(b). Let $B = [a, a + h)$ with $a, h \in W\mathbb{Z}$, $h > 0$, put $E = \{X \in B\}$, $F = \{Y \in B\}$, $I = F \setminus E$, $O = E \setminus F$, and let $\pi_\ell^B = \mathbb{P}(L = \ell \mid E)$. We first identify the edge events.

Lemma S8. *$I = \bigcup_{\ell\in\mathcal{L}}\{L = \ell,\ a - \ell/2 \le X < a\}$ and $O = \bigcup_{\ell\in\mathcal{L}}\{L = \ell,\ a + h - \ell/2 \le X < a + h\}$. On I, $Y = a$; on O, $Y = a + h$. In particular $Y \in W\mathbb{Z}$, $G = W$ and $S(G) = \mathcal{L}$ on $I \cup O$, and*

$$\mathbb{P}(I) = \sum_{\ell\in\mathcal{L}} \int_{a-\ell/2}^{a} h_\ell, \qquad \mathbb{P}(O) = \sum_{\ell\in\mathcal{L}} \int_{a+h-\ell/2}^{a+h} h_\ell, \qquad h_\ell = g\,\pi_\ell.$$

Proof. Fix ℓ and work on $\{L = \ell\}$. Since $\ell \mid W$, a/ℓ and $(a + h)/\ell$ are integers, and $K_\ell(x) = \lfloor x/\ell + \frac{1}{2} \rfloor$ is non-decreasing. If $x \ge a$ then $K_\ell \ge a/\ell$ and $Y \ge a$; if $x \ge a + h$ then $Y \ge a + h$; if $x < a$ then $K_\ell \le a/\ell$, so $Y \le a$, with $Y = a$ exactly when $x/\ell + \frac{1}{2} \ge a/\ell$, that is $x \ge a - \ell/2$. Hence $Y \in B$ with $X \notin B$ occurs exactly when $a - \ell/2 \le x < a$, and then $Y = a$. Likewise $X \in B$ with $Y \notin B$ occurs exactly when $Y \ge a + h$, that is $a + h - \ell/2 \le x < a + h$, and then $Y = a + h$. The probabilities follow from $\mathbb{P}(L = \ell, X \in dx) = h_\ell(x)\,dx$. □

Suppose Assumption 1 holds conditionally on E. The argument of Proposition 1 then gives $\mathbb{P}(G = d \mid L = \ell, E) = \Lambda_{d\ell}$, hence $\mathbb{P}(G = d, E) = \mathbb{P}(E)(\Lambda\pi^B)_d$. Writing $F = (E \setminus O) \cup I$

and using Lemma S8,

$$\mathbb{P}(G=d,F)=\mathbb{P}(E)(\Lambda\pi^B)_d-\mathbb{P}(O)\mathbf{1}\{d=W\}+\mathbb{P}(I)\mathbf{1}\{d=W\},\quad \mathbb{P}(F)=\mathbb{P}(E)+\mathbb{P}(I)-\mathbb{P}(O).$$

Let $p_d^F=\mathbb{P}(G=d\mid F)$ be the class distribution of the reports in the band and $\Delta=\mathbb{P}(I)-\mathbb{P}(O)$ the net flow into the band. Then

$$p^F=\frac{\mathbb{P}(E)\,\Lambda\pi^B+\Delta\,e_W}{\mathbb{P}(F)},\qquad p^F-\Lambda\pi^B=\frac{\Delta}{\mathbb{P}(F)}\big(e_W-\Lambda\pi^B\big),\tag{S5}$$

where e_W is the unit vector of the class $d=W$. Summing over $d\in D_S$ gives the same identities for the pattern distribution, with B in place of Λ and $e_{\mathcal{L}}$ in place of e_W (the pattern $\mathcal{L}=S(W)$ has $D_{\mathcal{L}}=\{W\}$). In every case $\sup_d|p_d^F-(\Lambda\pi^B)_d|\le|\Delta|/\mathbb{P}(Y\in B)$, and the same holds for the pattern distribution.

The weights when $V=J$. By Lemma S7, $V=J$ means $\mathcal{L}$ is closed under lcm, so $W=\operatorname{lcm}\mathcal{L}\in\mathcal{L}$ is the coarsest level. Then $B_{\mathcal{L},\ell}=c_{\mathcal{L}}\ell=\ell/W$ and the column of B indexed by the level W is $e_{\mathcal{L}}$, because $\ell=W$ divides no other attainable lcm than W itself. With B invertible, (S5) reads $p_S^F=B\pi^F$ for

$$\pi^F=\frac{\mathbb{P}(E)\,\pi^B+\Delta\,e_W}{\mathbb{P}(F)},\qquad \pi^F-\pi^B=\frac{\Delta}{\mathbb{P}(Y\in B)}\big(e_W-\pi^B\big),\tag{S6}$$

e_W now the unit vector of the level W. By Proposition 2 and the consistency of the maximum-likelihood estimator, π^F is the population value of the mixture weights computed from the reports in B whenever it lies in the simplex, that is whenever $\mathbb{P}(E)\pi_W^B\ge\mathbb{P}(O)-\mathbb{P}(I)$; it always satisfies $\mathbf{1}^\top\pi^F=1$. Under that condition,

$$\max_\ell|\pi_\ell^F-\pi_\ell^B|=\frac{|\Delta|}{\mathbb{P}(Y\in B)}\max\big(1-\pi_W^B,\ \max_{\ell\neq W}\pi_\ell^B\big)\le\frac{|\mathbb{P}(I)-\mathbb{P}(O)|}{\mathbb{P}(Y\in B)}.\tag{S7}$$

The error is thus governed by the net, not the gross, flow across the edges, and only by the half-neighbourhoods $[a-\ell/2,a)$ and $[a+h-\ell/2,a+h)$ on the left of each edge, the band being closed on the left.

The stated bound. Let $\beta=\mathbb{P}(X\in B)^{-1}\sum_{\ell\in\mathcal{L}}\ell\sup_{x\in\partial B\pm\ell/2}h_\ell(x)$ be the expression in the proposition, the supremum taken over $[a-\ell/2,a+\ell/2]\cup[a+h-\ell/2,a+h+\ell/2]$. By Lemma S8, $\mathbb{P}(I)\le\sum_\ell(\ell/2)\sup_{[a-\ell/2,a]}h_\ell\le\beta\mathbb{P}(E)/2$ and likewise $\mathbb{P}(O)\le\beta\mathbb{P}(E)/2$. If $\mathbb{P}(I)\ge\mathbb{P}(O)$ then $\mathbb{P}(F)\ge\mathbb{P}(E)$ and the right-hand side of (S7) is at most $\mathbb{P}(I)/\mathbb{P}(E)\le\beta/2$. If $\mathbb{P}(O)>\mathbb{P}(I)$, put $x=\mathbb{P}(O)-\mathbb{P}(I)\le\beta\mathbb{P}(E)/2$; then $\mathbb{P}(F)=\mathbb{P}(E)-x$ and the right-hand side is $x/(\mathbb{P}(E)-x)\le(\beta/2)/(1-\beta/2)=\beta/(2-\beta)\le\beta$ provided $\beta\le1$. When $\beta>1$ the

bound β holds trivially, the left-hand side of (S7) being a difference of probabilities. This proves part (b) as stated, with the sharper form (S6)–(S7). □

The case $V > J$. Then $W \notin \mathcal{L}$, the pattern $\mathcal{L}$ is not among the rows $S(\ell)$, $\ell \in \mathcal{L}$, and $e_{\mathcal{L}} \notin \operatorname{col} B$; the class distribution of the reports in the band is not exactly a model distribution, and the fitted weights depend on the fitting criterion. The triangular fit that solves (S4) on the rows $S(\ell)$, $\ell \in \mathcal{L}$, returns $\mathbb{P}(E)\pi^B/\mathbb{P}(F)$ exactly, since those rows of $e_{\mathcal{L}}$ vanish, so its error is $|\Delta|\,\pi^B_\ell/\mathbb{P}(F)$ and obeys (S7). The maximum-likelihood fit is a continuous function of the pattern distribution and its error is of the same order $|\Delta|/\mathbb{P}(F)$, but we do not give an explicit constant for it. Appending W to the candidate ladder, which is harmless when no respondent rounds to W, restores $V = J$ for the ladders of the paper.

S6 Additional simulation results

The designs and mechanisms are those of Section 6. Rejection frequencies are at the five per cent level. Tables S1 and S2 use 200 replications per cell, for which a frequency of 0.05 has a Monte Carlo standard error of about 0.015; Tables S3–S5 use 400 replications for unimodal and 200 for bimodal designs, as in Table 1 (standard errors of about 0.011 and 0.015 at 0.05); Table S6 uses 60.

References

Marcus Groß and Ulrich Rendtel. Kernel density estimation for heaped data. *Journal of Survey Statistics and Methodology*, 4(3):339–361, 2016. doi: 10.1093/jssam/smw011.

Table S1: Specification test of Theorem 1: rejection frequencies. "Correct ladder": the test on every candidate level dividing the certified width. "Decimal ladder": the same test on the levels $1, 10, 100, \ldots$ only, which omits the active levels 500, 2500 and 5000. "Below": average share of reports below W.

			correct ladder			decimal ladder		
mechanism	design	below	2,000	5,000	20,000	2,000	5,000	20,000
ACS	LN	0.04	0.06	0.07	0.04	1.00	1.00	1.00
	GA	0.04	0.04	0.05	0.07	1.00	1.00	1.00
	SM	0.03	0.05	0.04	0.04	1.00	1.00	1.00
	SHLD	0.01	0.03	0.08	0.09	1.00	1.00	1.00
	PLAT	0.00	0.07	0.05	0.03	1.00	1.00	1.00
	FLATDEC	0.02	0.04	0.06	0.06	1.00	1.00	1.00
	FLATTOP	0.01	0.03	0.05	0.06	1.00	1.00	1.00
types	LN	0.05	0.05	0.04	0.05	1.00	1.00	1.00
	GA	0.05	0.06	0.07	0.06	1.00	1.00	1.00
	SM	0.03	0.04	0.06	0.04	1.00	1.00	1.00
	SHLD	0.01	0.05	0.07	0.08	1.00	1.00	1.00
	PLAT	0.00	0.05	0.04	0.04	1.00	1.00	1.00
	FLATDEC	0.02	0.04	0.05	0.05	1.00	1.00	1.00
	FLATTOP	0.01	0.07	0.04	0.04	1.00	1.00	1.00
ACS + \$25k	LN	0.54	0.99	1.00	1.00	1.00	1.00	1.00
	GA	0.53	0.69	0.99	1.00	1.00	1.00	1.00
	SM	0.02	0.05	0.04	0.07	0.88	0.96	1.00
	SHLD	0.55	1.00	1.00	1.00	1.00	1.00	1.00
	PLAT	0.41	1.00	1.00	1.00	1.00	1.00	1.00
	FLATDEC	0.37	0.78	1.00	1.00	1.00	1.00	1.00
	FLATTOP	0.23	0.05	0.08	0.30	1.00	1.00	1.00

Table S2: Calibration of the activity cut-off. The ACS mechanism with an additional level of \$25,000 used with probability q; the cell test is run at $w =$ \$10,000, which ignores that level. Rejection frequencies of a true null. On the designs not shown (LN, GA, SM, SHLD, PLAT) every frequency is at most 0.01 for $q \leq 0.05$.

design	n	$q = 0$	0.01	0.02	0.03	0.05	0.10
FLATDEC	2,000	0.02	0.02	0.01	0.02	0.05	0.12
	10,000	0.02	0.02	0.01	0.04	0.09	0.94
	50,000	0.01	0.01	0.09	0.22	0.87	1.00
FLATTOP	2,000	0.01	0.02	0.02	0.04	0.08	0.23
	10,000	0.03	0.02	0.04	0.08	0.35	0.97
	50,000	0.02	0.04	0.22	0.46	0.95	1.00

Table S3: Rejection frequencies, $n = 10{,}000$, exact and type-dependent reporting. Columns and conventions as in Table 1.

	exact reports						types					
design	EM	dip	floor	jit.	rer.	cell	EM	dip	floor	jit.	rer.	cell
LN	.04	.03	.07	.03	.00	.00	1.00	1.00	.03	.03	.06	.00
GA	.06	.05	.20	.06	.00	.00	1.00	1.00	.27	.07	.00	.00
SM	.05	.03	.21	.04	.00	.00	1.00	1.00	.16	.03	.07	.00
SHLD†	.20	.42	.41	.19	.00	.00	1.00	1.00	.28	.22	.04	.00
PLAT	.04	.35	.07	.04	.00	.00	1.00	1.00	.06	.03	.00	.00
FLATDEC†	1.00	1.00	.27	1.00	.01	.05	1.00	1.00	.30	1.00	.00	.02
FLATTOP	.01	.01	.23	.01	.01	.01	1.00	1.00	.22	.02	.00	.01
N1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
N2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
N3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
M1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M4	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M5	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
S1	1.00	1.00	1.00	1.00	.83	.65	1.00	1.00	1.00	1.00	.84	.74
S2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table S4: Rejection frequencies, $n = 2{,}000$, ACS mechanisms. Columns and conventions as in Table 1.

	ACS mechanism						ACS + \$25k					
design	EM	dip	floor	jit.	rer.	cell	EM	dip	floor	jit.	rer.	cell
LN	1.00	1.00	.16	.04	.00	.00	1.00	1.00	.02	.07	.01	.00
GA	1.00	1.00	.22	.04	.00	.00	1.00	1.00	.10	.05	.19	.00
SM	1.00	1.00	.19	.03	.03	.00	1.00	1.00	.21	1.00	.02	.00
SHLD†	1.00	1.00	.36	.14	.01	.00	1.00	1.00	.00	.17	.01	.00
PLAT	1.00	1.00	.26	.06	.00	.00	1.00	1.00	.00	.04	.21	.00
FLATDEC†	1.00	1.00	.41	.75	.00	.01	1.00	1.00	.39	.65	.00	.00
FLATTOP	1.00	1.00	.28	.02	.00	.01	1.00	1.00	.20	.03	.01	.00
N1	1.00	1.00	1.00	1.00	1.00	.00	1.00	1.00	.22	1.00	1.00	.00
N2	1.00	1.00	.89	1.00	1.00	.00	1.00	1.00	.00	1.00	.77	.00
N3	1.00	1.00	1.00	1.00	1.00	.00	1.00	1.00	1.00	1.00	1.00	.00
M1	1.00	1.00	1.00	1.00	.97	.89	1.00	1.00	1.00	1.00	1.00	.00
M2	1.00	1.00	1.00	1.00	.94	.90	1.00	1.00	1.00	1.00	1.00	.00
M3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.96	.00
M4	1.00	1.00	1.00	1.00	.65	1.00	1.00	1.00	.02	.96	.02	.00
M5	1.00	1.00	1.00	1.00	.00	.82	1.00	1.00	.01	1.00	.03	.00
M6	1.00	1.00	1.00	1.00	.98	1.00	1.00	1.00	1.00	1.00	.65	.85
S1	1.00	1.00	1.00	1.00	.17	.14	1.00	1.00	.00	.96	.41	.00
S2	1.00	1.00	1.00	1.00	.86	.72	1.00	1.00	.01	1.00	.29	.00

Table S5: Rejection frequencies, $n = 2{,}000$, exact and type-dependent reporting. Columns and conventions as in Table 1.

	exact reports						types					
design	EM	dip	floor	jit.	rer.	cell	EM	dip	floor	jit.	rer.	cell
LN	.03	.03	.05	.02	.00	.00	1.00	1.00	.08	.03	.02	.00
GA	.06	.04	.19	.05	.00	.00	1.00	1.00	.13	.06	.01	.00
SM	.05	.03	.20	.05	.01	.00	1.00	1.00	.15	.04	.01	.00
SHLD†	.14	.27	.35	.14	.00	.00	1.00	1.00	.33	.14	.00	.00
PLAT	.04	.30	.13	.04	.00	.00	1.00	1.00	.17	.05	.00	.00
FLATDEC†	.84	.91	.26	.84	.01	.07	1.00	1.00	.36	.79	.00	.03
FLATTOP	.04	.03	.23	.04	.02	.00	1.00	1.00	.25	.04	.01	.02
N1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
N2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.97	1.00	1.00	.00
N3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.00
M1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M2	1.00	1.00	1.00	1.00	1.00	.99	1.00	1.00	1.00	1.00	.99	.98
M3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
M4	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.98	1.00
M5	1.00	1.00	1.00	1.00	.95	1.00	1.00	1.00	1.00	1.00	.78	.85
M6	1.00	1.00	1.00	1.00	.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
S1	1.00	1.00	1.00	1.00	.20	.17	1.00	1.00	1.00	1.00	.23	.14
S2	1.00	1.00	1.00	1.00	.95	.90	1.00	1.00	1.00	1.00	.93	.81

Table S6: The inverted test of Section 5 at $w = \$5{,}000$, ladder $\{1, 100, 1000, 5000, 10^4\}$, under a profile that moves strongly with income; rejection frequencies at the five per cent level over 60 replications, 99 bootstrap draws. "True": the latent-conditional profile used to generate the data. "Band": the band estimator of Proposition S1(b) with bands of \$80,000 and \$160,000. "Logit": a multinomial logit of the level on log income fitted by EM to the reports. "Marginal": the constant profile $\hat{\bar{\pi}}$. "Omit": the true profile with the \$5,000 level removed from the ladder (a separate run of 60 replications, in which the true profile gave rejection rates of at most 0.02 on every design shown). M1 is bimodal. The simulations use cells $[(k-1)w, kw)$ and the exact reporting map.

design	n	true	band 80k	band 160k	logit	marginal	omit
LN	50,000	0.00	0.62	1.00	0.07	1.00	0.97
	200,000	0.00	1.00	1.00	0.27	1.00	1.00
GA	50,000	0.00	1.00	1.00	0.03	1.00	0.95
	200,000	0.00	1.00	1.00	0.22	1.00	1.00
PLAT	50,000	0.00	0.30	1.00	0.18	1.00	1.00
	200,000	0.03	0.63	1.00	0.40	1.00	1.00
SHLD	50,000	–	–	–	–	–	1.00
	200,000	–	–	–	–	–	1.00
SM	50,000	–	–	–	–	–	0.00
	200,000	–	–	–	–	–	0.00
M1	50,000	1.00	1.00	1.00	1.00	1.00	–
	200,000	1.00	1.00	1.00	1.00	1.00	–